



Using Predator Gut Content eDNA to Improve Detection of California Tiger Salamanders in Seasonal Wetlands

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Introduction

Detecting California Tiger Salamanders (*Ambystoma californiense*, CTS) is challenging due to their cryptic life stages and the variable conditions of seasonal wetlands. Traditional survey methods, such as larval seining, can be labor-intensive and unreliable, while environmental DNA (eDNA) approaches, though promising, may still produce inconsistent results. Predatory invertebrates that consume CTS larvae may retain detectable prey DNA, offering a novel opportunity to use predator gut contents as a “biological sampler.” This study explores whether predator-derived eDNA can improve detection of CTS in seasonal wetland systems.

California Tiger Salamander Ecology

- Federally threatened amphibian
- Breeds in seasonal wetlands
- Larval survival depends on hydroperiod
- Cryptic and difficult to detect
- Preyed upon by aquatic invertebrates
- Predator-prey interactions enable direct detection



Research Questions

1. Can predator gut contents be used to detect California Tiger Salamander (CTS) DNA?
2. Do predator-based detections align with traditional larval survey methods?
3. How does detection success vary across different predator taxa?
4. How do environmental conditions influence detection probability?
5. Can predator gut DNA improve overall detection reliability in seasonal wetlands?

Conceptual Framework:

Predator larvae eats CTS



CTS DNA remains in gut



Predator larvae collected

Extracted gut content analyzed via PCR.

Predator gut DNA acts as a “biological sampler”



Methods

Field Sampling

- 10 ponds at MVPGR
- SWAMP dip-net transects
- Individual specimen collection storage

Lab Processing

- Ethanol preservation
- Cold storage
- Sterile dissection

Molecular Analysis

- Species-specific PCR
- Triplicate runs (2 of 3 rule)

